# Freight Rate Forecasting

Kipngeno Koech · Machine Learning Engineer assessment · Code: github.com/kkipngenokoech/freight-rate-forecasting

## 1. Train/test split and validation approach

The labelled data (`train_test.csv`, 48,000 loads) covers **Jan–Oct 2025**. The 12,000 loads to predict cover **Nov–Dec 2025**. This is a two-month-ahead forecast, so a random train/test split would leak future market conditions into training and overstate accuracy. Every split below is by time.

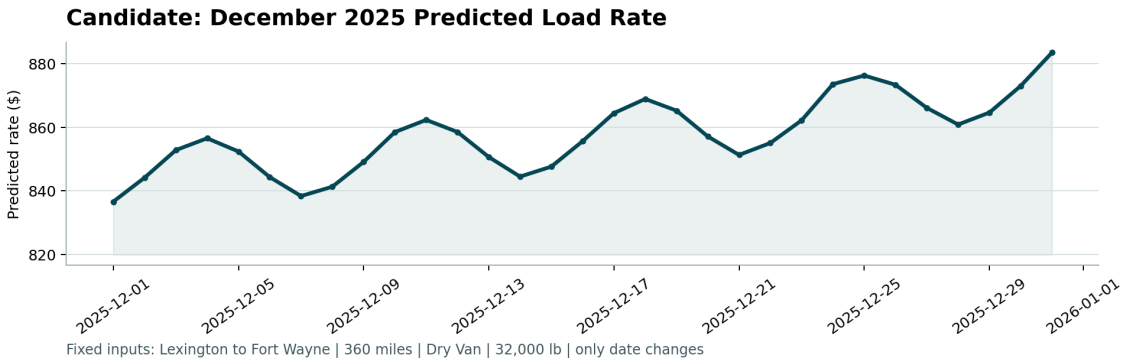
- **Primary holdout:** train on Jan–Aug, test on Sep–Oct. This is the same two-month horizon as the real task, and it includes a quarter end, as December does.
- **Rolling-origin backtest:** for each month from Jun to Oct, train on all earlier months and test on that month. This checks that the result is stable and not one lucky window.
- **Unseen-city check:** 8 cities in Nov–Dec never appear in training (1,447 loads). To mimic this, 8 cities are removed from the holdout's training set and those loads are scored separately.
- **Forecast-period conditions:** features that behave differently in Nov–Dec (see `quote_signal`, section 3) are scored with that behaviour simulated, not as they look in training.
- **Label noise:** about 1.4% of labels look corrupted. They are flagged using only each split's training window, so no holdout information leaks. Results are reported on clean rows and on all rows.

| Primary holdout (train Jan–Aug, test Sep–Oct) | MAPE clean | MAE clean | MAPE all rows |
|---|---|---|---|
| Baseline: median $/mile by equipment × distance band | 3.82% | $82.8 | 6.32% |
| Ridge regression on log $/mile | 2.59% | $54.6 | 5.11% |
| Gradient boosting alone | 3.65% | $86.8 | 6.06% |
| **Ridge + gradient boosting hybrid (final)** | **1.34%** | **$31.7** | **3.87%** |

| Rolling backtest, MAPE on clean rows | Jun | Jul | Aug | Sep | Oct | Mean |
|---|---|---|---|---|---|---|
| Ridge | 2.52% | 2.58% | 2.58% | 2.53% | 2.65% | 2.57% |
| Gradient boosting alone | 2.31% | 2.25% | 3.05% | 2.07% | 5.13% | 2.96% |
| **Hybrid (final)** | **1.27%** | **1.39%** | **1.32%** | **1.21%** | **1.48%** | **1.33%** |

In the unseen-city check, loads touching a held-out city score **1.61%**, against 1.35% for the other loads.

## 2. December prediction chart




Produced by the provided `score.py`. Both output files passed its validation (12,000 final predictions, 31 December rows). Lane: Lexington → Fort Wayne, 360 mi, Dry Van, 32,000 lb. The prediction follows the weekly market cycle and rises from about $837 to $883 into the quarter end. Historical rates on this lane ranged from $758 to $934 (mean $823, 21 loads).

# Additional notes

## 3. Data findings and quality issues

- **`quote_signal` leaks the label in training but is noise in the forecast period.** Across the whole dataset it looks useless (correlation 0.05). Within a single day it tracks $/mile closely (median |corr| 0.63), and the sign of that relationship flips every few weeks. The relationship is absent in August (0.05) and across all of Nov–Dec. A model that keeps it scores 1.35% on the holdout; with `quote_signal` shuffled within each day to mimic Nov–Dec, that rises to 1.51%. The feature is dropped.
- **Suspected corrupted labels (677 rows, 1.41%):** rates at 0.17–0.45× or 2.2–5.4× the expected rate, with an empty gap between them and normal rows. They are flagged with a first-pass model and excluded from training.
- **Negative weights** (292 train, 145 forecast): the sign is flipped, since these rows price like normal loads. **Weights capped at 47,500 lb** (1,191 rows): the value is kept and a cap flag is added. **Missing weight** (300) is left for the model to handle.
- **Missing `market_index`** (374): filled with that day's mean, because the index is a daily market level plus small noise.
- **Trend and seasonality:** rates drift up about +0.6% per month beyond what `market_index` explains, and they ramp up over the 5 weeks before each quarter end (most for flatbed, least for dry van).

## 4. Model

The model predicts log $/mile in two stages. A **ridge regression** captures effects that must extend past the training dates: distance curve, market index, equipment, weight, time trend, city effects, and a quarter-end ramp per equipment type. **Gradient boosting** then fits the ridge's residuals using geography, daily market level, day of week, days to quarter end and a shrunk per-lane effect. It gets no raw date, so the trend stays in the ridge. Boosting alone cannot extrapolate the trend: see its 5.13% in October. Loads touching a city not seen in training are routed to a twin model without city effects.

## 5. Limitations

- The time trend is extended two months beyond the data.
- The month-to-month level, which jumps at each quarter start, is the largest remaining source of error.
- One year of data cannot capture holiday effects in late December.